

Rayyan Abhram

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Portfolio: www.rayyanabhram.com

Skills

Programming Languages: C, C++, Python, RUST, VHDL

Software: Altium designer, KiCad, Vivado, Fusion 360, OnShape, Creo, VS-Code, GitHub

Hardware & Tools: ESP32, NRF52, STM32 SoCs, SMD Reflow soldering, SWD/JTAG

Prototyping: FDM / SLA 3D printing, CNC machining, Silicone casting, Vacuum casting

IT: Microsoft office, LaTeX, MATLAB

Education

Queen Mary University of London – 1st Class BEng Robotics Engineering June 2025

Plovdiv Medical University – Diploma: Dental Medicine May 2022

Norlington School & Sixth form – *A Levels*: Biology (A), Math (B), Chemistry (B) June 2020

Experience

Junior Hardware Engineer, Timebeat – London, UK Feb 2026 – Present

- Contributing to the development of the open-timcard platform for nano-second and pico-second timing; supporting schematic updates and multi-layer PCB routing in Altium designer
- Designed and implemented a power-delivery network on the flagship open timcard, introducing USB-C power delivery and power over ethernet (PoE)
- Routed controlled-impedance differential traces & high speed nets for controlled RF delivery, antenna splitting paths and timing functionality
- Collaborating with FPGA consultants during custom board bring-up, system debug, and SFP-based master-slave timing validation

Software Engineer, Cedar Labs – London, UK Sept 2025 – Dec 2025

- Built scalable web-scraping systems to extract structured product data from Amazon and Shopify stores for pricing, positioning, and competitor analysis
- Developed automated pipelines to collect, clean, and normalize product metadata (pricing, reviews, ratings, descriptions, stock levels) for downstream analytics
- Designed dashboards to visualize competitive trends, price movements, review sentiment, and category performance for client decision-making
- Implemented AI-driven analysis tools to generate structured feedback on product listings (SEO quality, description clarity, review sentiment, differentiation gaps)

Hardware Engineer Intern, Fiddlie Technologies – London, UK May 2024 – Aug 2025

- Engineered multilayer (Rigid & flex) PCBs in Altium designer for project specific applications, with readable documentation and BoM optimised boards cutting costs up to 15%.
- Evaluated the performance of hand assembled prototypes and validated hardware designs using oscilloscopes, power profilers and DMMs.
- Implemented real-time firmware in Zephyr/FreeRTOS with I2C, SPI, and UART support, enabling external device integration, LVGL compatible interfaces and low power consumption, achieving sub-10 μ A current consumption.
- Fabricated and assembled hardware using Fusion 360 and CircuitMind, outputting components through CNC machining and SLA/FDM 3D printing, producing functional enclosures in <24 hours for rapid iteration.

Low Voltage Systems Engineer, QMFS – London, UK Sept 2023 – Jun 2024

- Senior member of QMFS electric vehicle department, directed the design of two distinct safety systems; placed 4th in a nationwide competition.
- Rear Lighting System: designed brake-light PCB using FET-switched LED array, translating driver braking status into clear visual signaling.
- Designed break system plausibility device using comparator based logic to detect and prevent simultaneous break and throttle engagement.

Research Associate, Multi-Modal human interaction Lab

Oct 2023 – May 2024

- Invented a novel EEG platform using soft-robotic actuators for adaptive electrode positioning using closed loop pneumatic control.
- Produced 20+ precision silicone actuators using Ecoflex 30 to achieve consistent and compliant pneumatic responses.
- Programmed a Python-based control system for sensor feedback, valve actuation, and real-time adaptation with minimal latency.

Founding Board Member, OMCUR

Oct 2023 – May 2024

- Launched the Queen Mary Center for undergraduate research as a founding member, establishing a cross faculty platform for student led research.
- Organized and hosted campus wide events, working in multidisciplinary student teams to promote undergraduate research to higher academic bodies through both presenting and hosting.
- Created promotional materials and spearheaded outreach campaigns, boosting student engagement and participation in research events to 80+ students within a single academic year.
- Forged strong relationships with partner institutions to foster center growth and collaboration to expand research opportunities across disciplines.

Technical Projects

NB: For more projects see: www.rayyanabhram.com